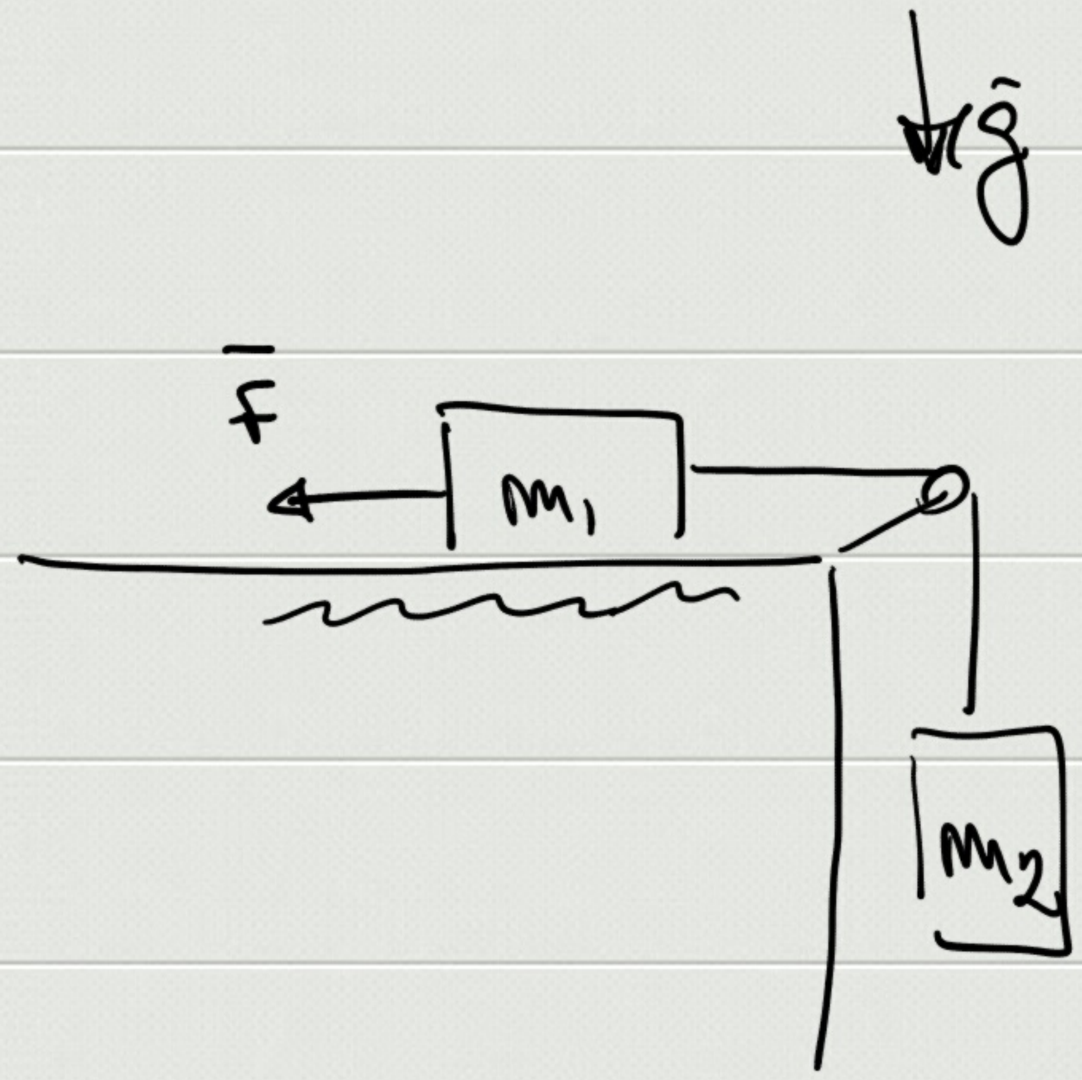


$m_1 = 10\ kg$

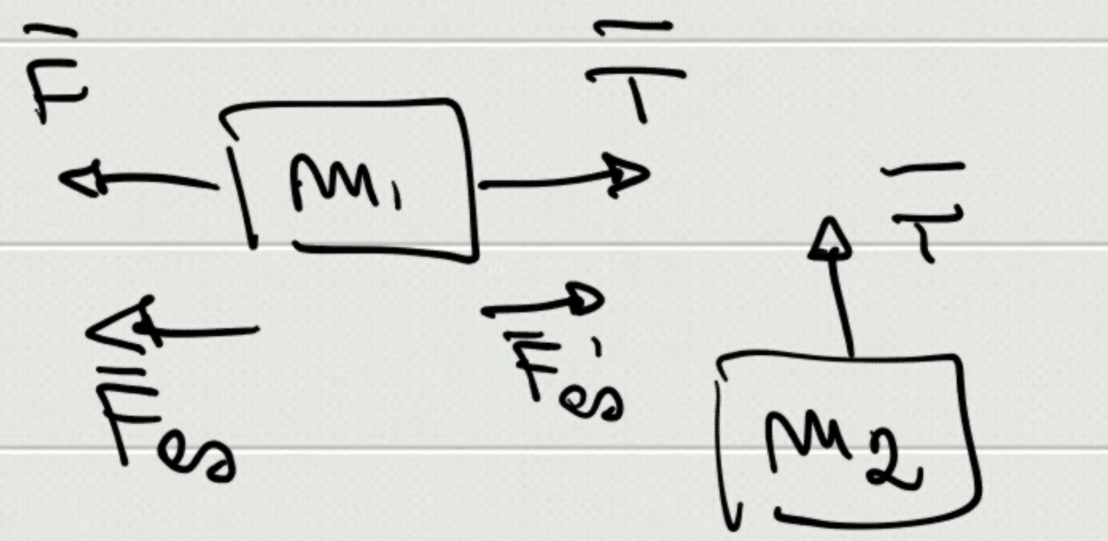
$m_2 = 5\ kg$

$\mu_s = \mu_d = 0.4$

Statico

$F = ?$; eq. statico

$$\begin{cases} \cancel{F = T} \\ F - T - F_{es} = 0 \quad * \\ F - T + F_{es} = 0 \quad * \\ F = \cancel{\mu_s m_1 g} \end{cases}$$



$\downarrow m_2\bar{g}$

$\Downarrow$

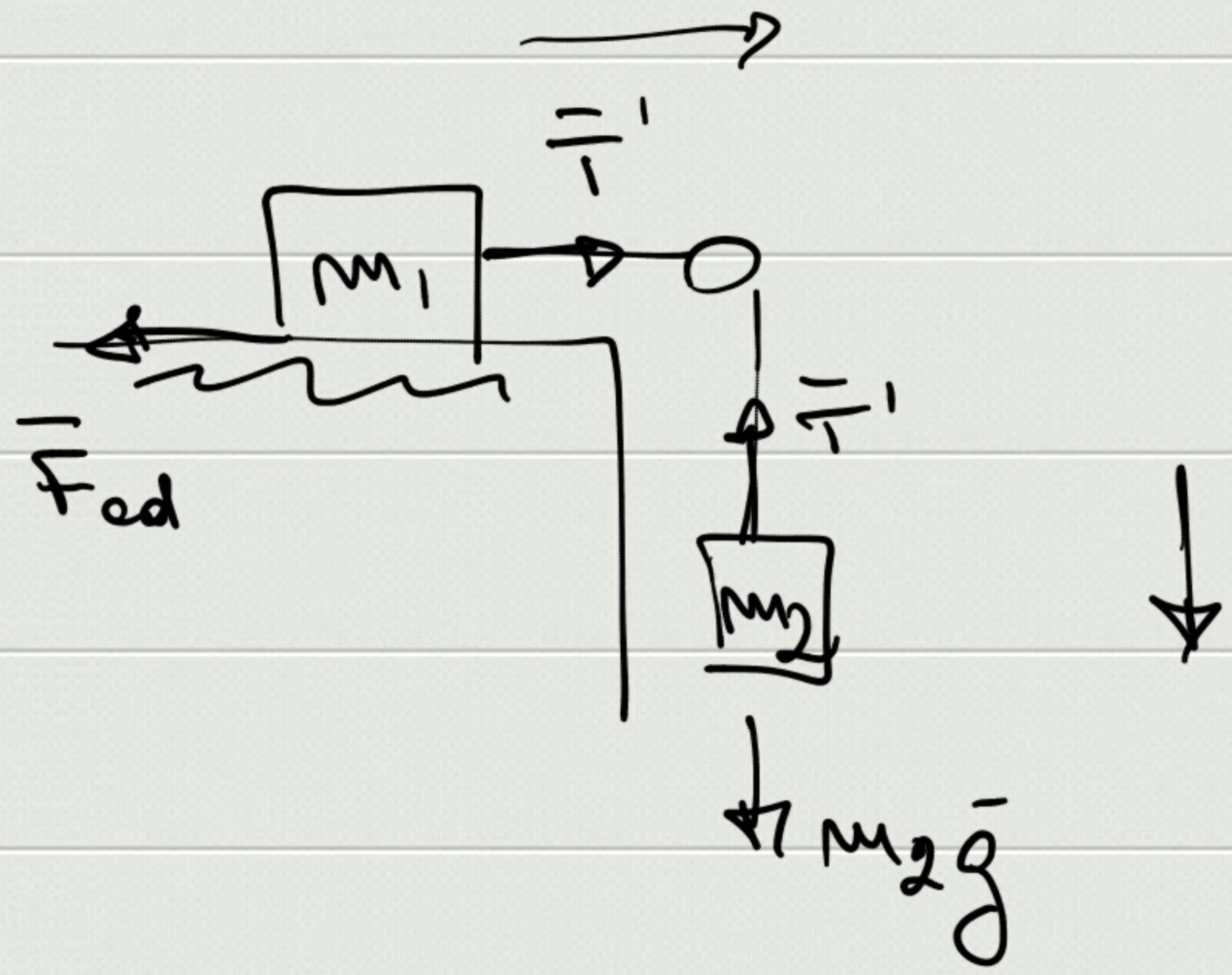
$T = m_2 g$

$$F_{es} = F - m_2 g \leq F_{es,max} = \mu_s m_1 g$$

$$F \leq (\mu_s m_1 + m_2) g = 88.2\ N \searrow F_{max}$$

$$F_{es} = m_2 g - F \leq F_{es,max} = \mu_s m_1 g$$

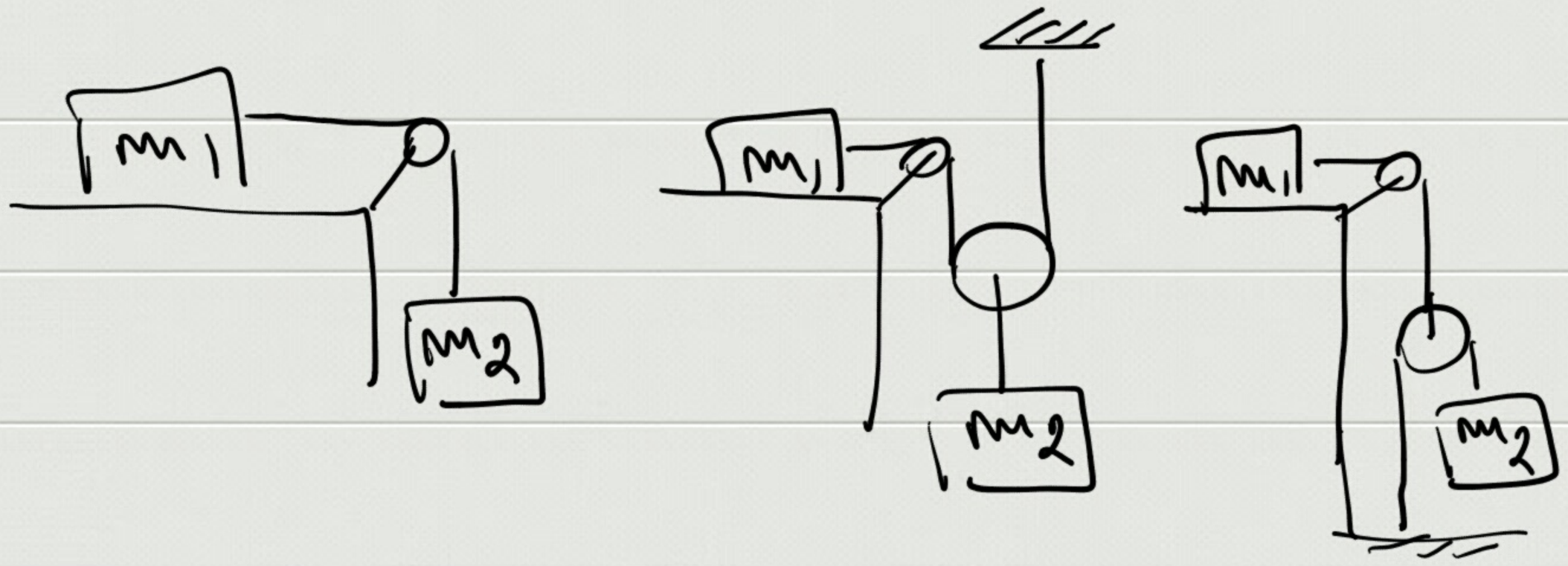
$$F \geq (m_2 - \mu_s m_1) g = 9.8\ N \nearrow F_{min}$$



$$\begin{cases} T' - \mu_d m_1 g = m_1 a \\ m_2 g - T' = m_2 a \end{cases}$$

$$(m_2 - \mu_d m_1) g = (m_1 + m_2) a \Rightarrow$$

$$a = \frac{m_2 - \mu_d m_1}{m_1 + m_2} = 0.65 \ m/s^2$$

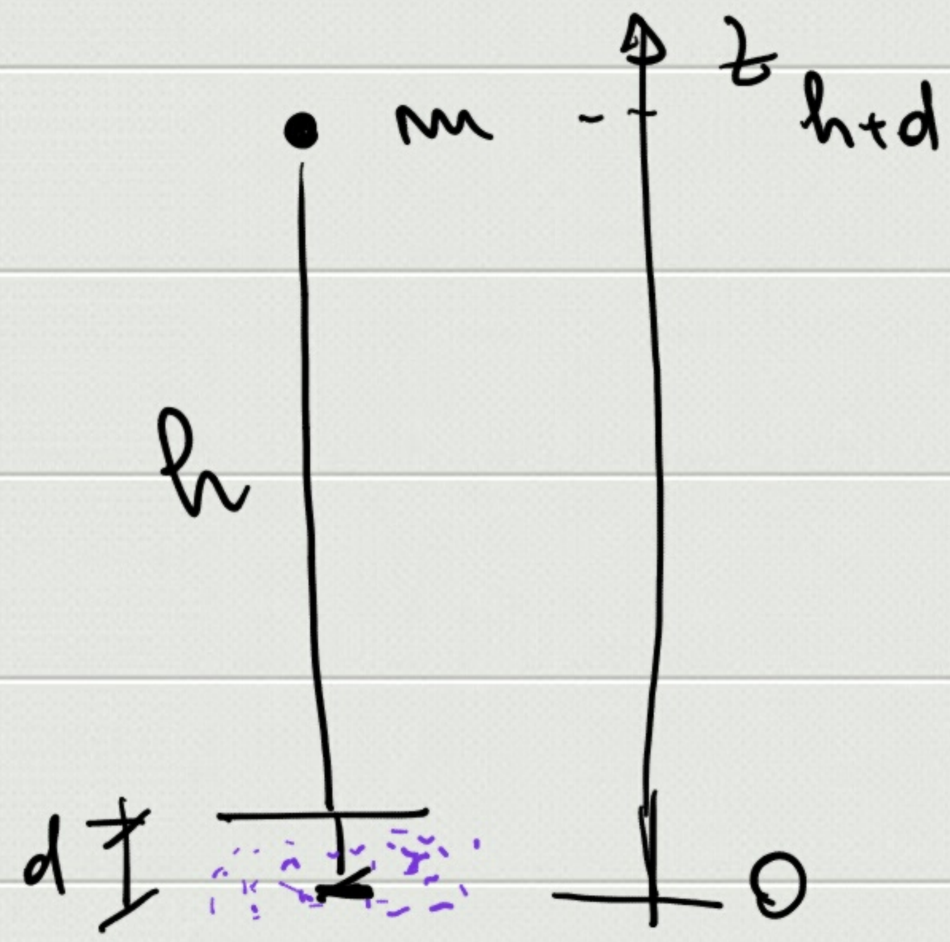


$a_1 = ?$ $\boxed{a_1 > g}$ $\left(\frac{m_2}{m_1}\right)_{min} = ?$

1) $a_1 = \frac{m_2}{m_1 + m_2} g$

2) $a_1 = \frac{2m_2}{4m_1 + m_2} g$ * $\left(\frac{m_2}{m_1} > 4\right)$

3) $a_1 = \frac{2m_2}{m_1 + 4m_2} g$



$m = 0.1\ kg$

$h = 3\ m$

$d = 0.03\ m$

$F_{sabbia} = cost = ?$

$$* - \begin{cases} v^2 = 2gh \\ 0 = v^2 + 2(g - a_s)d \end{cases}$$

$\downarrow z$

$v_f^2 = v_i^2 + 2a\Delta x$

$$* - \quad -F_s d = -mg(h+d)$$

$W_{nc} = \Delta E_m \ [(h+d)]$

$$* - \begin{cases} mgh = \frac{1}{2}mv^2 \\ -F_s d = -\left(mgd + \frac{1}{2}mv^2\right) \end{cases}$$

$E_m = cost$

$W_{nc} = \Delta E_m \ [d]$

$$* - \quad W_{peso} + W_{F_s} = 0$$

$W_{TOT} = \Delta E_K$

$$0 = \cancel{2}gh + \cancel{2}gd - \cancel{2}a_s d \Rightarrow a_s = g\frac{h+d}{d}$$

$$F_s = m a_s = mg\frac{h+d}{d} = 98\ N$$

$$-\Delta E_{p,peso} + W_{F_s} = 0 \Rightarrow -\left(\emptyset - mg(h+d)\right) - F_s d = 0$$